DS	**Session**	Linked List	**Question Information**	Level 1 Challenge 36

Question description

A long time ago, there was a desolate village in India. The ancient buildings, streets, and businesses were deserted. The windows were open, and the stairwell was in disarray. You can be sure that it will be a fantastic area for mice to romp about in! People in the community have now chosen to provide high-quality education to young people in order to further the village's growth.

As a result, they established a programming language coaching centre. People from the coaching centre are presently performing a test. Create a programme for the GetNth() function, which accepts a linked list and an integer index and returns the data value contained in the node at that index position.

Example

Input: 1->10->30->14, index = 2
Output: 30
The node at index 2 is 30

Constraints

1< N < 1000

1< X < 1000

1 < I < 1000

Input Format
First line contains the number of datas- N.
Second line contains N integers(the given linked list).
Third Line Index I

Output Format

First Line indicates the linked list

second line indicates the node at indexing position

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
4
1 2 3 4
2
```

EXPECTED OUTPUT

```
Linked list:-->4-->3-->2-->1
Node at index=2:3
```

Test Case 2

INPUT (STDIN)

```
4
11 21 31 41
3
```

EXPECTED OUTPUT

```
Linked list:-->41-->31-->21-->11
Node at index=3:21
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
struct node
```

Test Case 2

KEYWORD

```
struct node *next;
```

Test Case 3

KEYWORD

```
int GetNth(struct node*
head,int index)
```

Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

CYCLOMATIC COMPLEXITY	TOKEN COUNT	NLOC
10	385	74

Code

solution.c

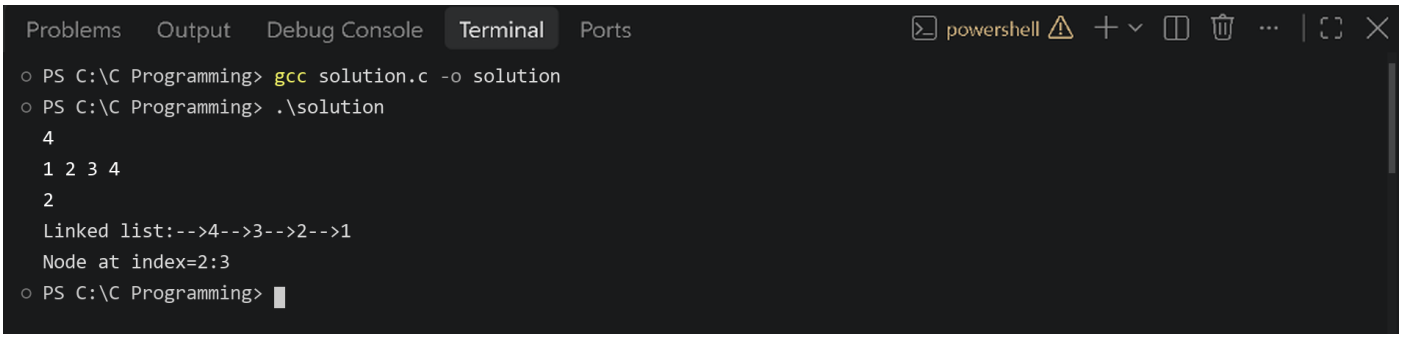
```c
#include <stdio.h>
#include <stdlib.h>

struct node {
    int data;
    struct node *next;
};

int GetNth(struct node* head, int index) {
    struct node* cur = head;
    int count = 1;
    while (cur) {
        if (count == index) return cur->data;
        count++;
        cur = cur->next;
    }
    return -1;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    struct node *head = NULL;
    for (int i = 0; i < n; i++) {
        int x; scanf("%d", &x);
        struct node *t = (struct node*)malloc(sizeof(struct node));
        t->data = x;
        t->next = head;
        head = t;
    }
    int idx;
    scanf("%d", &idx);
    printf("Linked list:");
    struct node *cur = head;
    while (cur) {
        printf("-->%d", cur->data);
        cur = cur->next;
    }
    printf("\nNode at index=%d:%d\n", idx, GetNth(head, idx));
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
4
1 2 3 4
2
Linked list:-->4-->3-->2-->1
Node at index=2:3
PS C:\C Programming>